

# CostVision

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What a part should cost to make — every number traced to a rate

**Designed & Developed by Avinash Bhosale**

Cost Engineering & Digital Innovation

**19 Processes**

**CAD to Cost, on Rules**

**AI Switched Off at JLR**

**DFM / DFA Advice**

**20 Regions**

**Assembly Roll-Up**

**Learning Curve**

**Supplier Quote Comparison**

**Windows Laptop, No Internet**

THE CHALLENGE

# Why CostVision? Costing by Hand Is Slow

Most should-costs are built by hand, one spreadsheet at a time. That is slow, and hard to defend.

**Slow**

A should-cost built by hand can take weeks. By the time it lands, the sourcing decision is often made.

**Quotes Taken on Trust**

Without our own number, a quote is hard to challenge. The supplier's margin stays hidden.

**No Common Method**

Each engineer builds the cost a different way. Assumptions are not written down, so numbers cannot be compared.

**Country Rates Are Hard**

Labour, machine, energy and material rates differ by country. Adjusting them by hand is easy to get wrong.

**Problems Found Late**

Cost problems show up near launch, not at concept. By then a change means new tooling and lost margin.

**No Floor Price**

Buyers go into a negotiation without a price they can defend line by line.

**~80%**

Industry rule of thumb, not a CostVision result: most of a part's cost is fixed at the design stage, yet most cost work happens after design freeze. We want a defensible number earlier.

THE SOLUTION

# What is CostVision?

A should-cost engine. Every number is arithmetic you can trace to a rate. It needs no AI.

**CAD to Cost**

Upload STEP, IGES or STL. The tool measures the part, and asks you where the geometry cannot decide.

**Form Entry**

Type the inputs for any of the 19 processes. Same engine, same eight cost buckets.

**19 Manufacturing Processes**

Machining to wiring harness, plus an assembly roll-up and an automotive software cost model.

**Rate Library**

Dated 16 June 2026: 328 materials, 178 machines, 42 labour grades. Editable and versioned.

**20 Regions**

Each with its own labour, energy and machine rates, shown in its own currency.

**Checks Its Own Work**

A self-audit and a P10–P90 uncertainty band come with every result.

**Supplier Quote Comparison**

Put a quote next to the should-cost, bucket by bucket, and see where the gap is.

**DFM / DFA Advice**

Rule-based advice on how to make the part cheaper. It never changes the price.

**Exports**

6-sheet Excel, PDF report and negotiation pack. Each matches the screen to the penny.

OPTIONAL AI MODE · SWITCHED OFF AT JLR

# AI Mode — Built In, Switched Off, Never Sets a Price

The AI code is in the product but turned off. If AI is ever approved, a setting turns it on — no rebuild.

### How It Works

**1. What it would do (if switched on)**

Read a plain-English description, a drawing or a PCB photo, and suggest the inputs: material, process route, features.

**2. What it never does**

Set a price. Every pound is calculated by the same rules engine, from the rate library.

**3. How it is kept in check**

Its suggestions are checked against the measured geometry. Every AI route is rate-limited per user.

**4. What JLR sees today**

Screens that need AI are hidden. If you reach one, the tool says "AI is switched off in this installation".

● **CostVision — JLR build, AI switched off**

You: Aluminium bracket, 6082-T6, 200 × 100 × 50 mm, three holes, 5,000 a year, UK

CostVision: AI is switched off in this installation.  
Costing, CAD geometry and the learning loop work without it.

- Upload the STEP file, or
- Fill in the machining form

You: Uploads bracket.step

CostVision: Measured: volume, weight, size, walls, holes.  
One question: which process route — machining or casting?  
Then: cost from the rate library, in eight buckets.

At JLR

No AI, no key

Same engine either way

Works offline

On by a setting

CAD TO COST

# CAD to Cost: Measured by the Tool, Checked by You

The geometry engine measures the part. Rules turn that into cost inputs. You answer what it cannot decide.

<0.01%

Engine vs hand calc,  
reference part

8

Cost buckets, each traced  
to a rate

6

Real parts pinned in  
regression

Supported formats

.STEP .STP .IGES .IGS .STL + .DXF flat pattern for sheet metal

Measured by the OpenCASCADE geometry kernel. The DXF gives the true blank instead of the bounding box.

What the Tool Measures

- Volume and weight — weight is volume × the density of the material you pick
- Size: bounding box, wall thickness and how much of the box the part fills
- Holes, bosses, threads and other features (from STEP and IGES)
- Face types and areas, for machining and coating
- A bottom-up CNC cycle estimate built from the feature table
- Walls over ~6 mm: offered casting, forging or machining — never sheet or moulding
- 13 of the 19 processes can be costed straight from CAD this way

Where the Tool Asks You — and What Comes Out

Process route

Asked if more than one route fits

Material family

Asked: steel, aluminium, plastic ...

Hole count on an STL

Asked: an STL has no feature table

Cycle time, machined STL

Costing blocked until you type one

Service questions

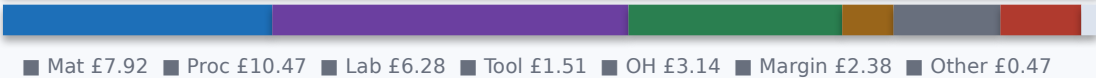
Pressure-tight? Safety-critical?

Sheet-metal blank

Measured from a DXF flat pattern

Real part: steering knuckle, forged steel, UK rates

£ 32.16



Pinned in the real-parts baseline. It records what the tool says — it has not been compared with a price JLR paid.

## 3D CAD VIEWER

# 3D CAD Viewer — Inspect, Measure and Check

Built into CostVision. STEP, IGES and STL in the browser — no separate CAD licence needed.

## Views & Navigation

Iso, front, top and right views, fit to screen, a view-cube, and a zoom that holds up on large parts. Full-window mode.

## Display & DFM Checks

Shaded or wireframe, bounding-box sizes, colour by face type, a colour per body with a legend, a draft and undercut map, and a wall-thickness map.

## Structure & Assembly

Model tree of bodies, features and faces. Section planes on X, Y and Z. Exploded view. Move or rotate one body while the rest stays put.

## Measure & Inspect

Distance, radius, diameter, angle and point position, read from the exact B-rep. Export to CSV, or snapshot into the report.

## Fits the Workspace

A dockable model tree, tool groups you can drag into your own order, and a toolbar that slides away. Works beside the app's sidebar.

## One viewer, many jobs

STEP / IGES / STL  
Runs in the browser  
Exact B-rep measurements  
No extra CAD licence

The same geometry  
feeds the cost engine →

CAD TO COST · MEASURED, NOT TYPED

# What the Tool Measures — and What It Feeds

Geometry is measured, not typed. Every £ is arithmetic you can trace to the rate library.

| Measured from the geometry (no typing)                 | → Feeds this cost driver                                    |
|--------------------------------------------------------|-------------------------------------------------------------|
| Volume — exact, from the geometry kernel (OpenCASCADE) | Material mass · stock size                                  |
| Weight = volume × density of the chosen material       | Raw-material £                                              |
| Bounding box / envelope                                | Machine sizing · stock · setups                             |
| Body / component count                                 | Assembly scope · BOM line count                             |
| Hole & boss table — Ø, depth, count (STEP / IGES)      | Secondary machining: drill / bore / tap                     |
| Wall thickness                                         | Which routes are offered · moulding and casting feasibility |
| Face types and areas (planar / cylindrical)            | Machining area · paint / coat area                          |
| CNC cycle-time and setup estimate                      | Process-time baseline for machining                         |
| DXF flat pattern (sheet metal, optional)               | True blank size, instead of the bounding box                |

## Worked example — Spur Gear (m3 · z38)

Case-hardening steel, ISO class 8 — measured:  
Volume 266.0 cm<sup>3</sup> · Steel weight 2.09 kg  
Envelope 120 × 120 × 30 mm · 1 body

→ Should-cost, UK rates (pinned in the baseline):  
Material £8.14 · Process £8.04 · Labour £3.61  
Tooling £0.96 · Overhead £2.49 · Margin £1.89  
+ packaging and logistics £0.36  
= £25.50 / part

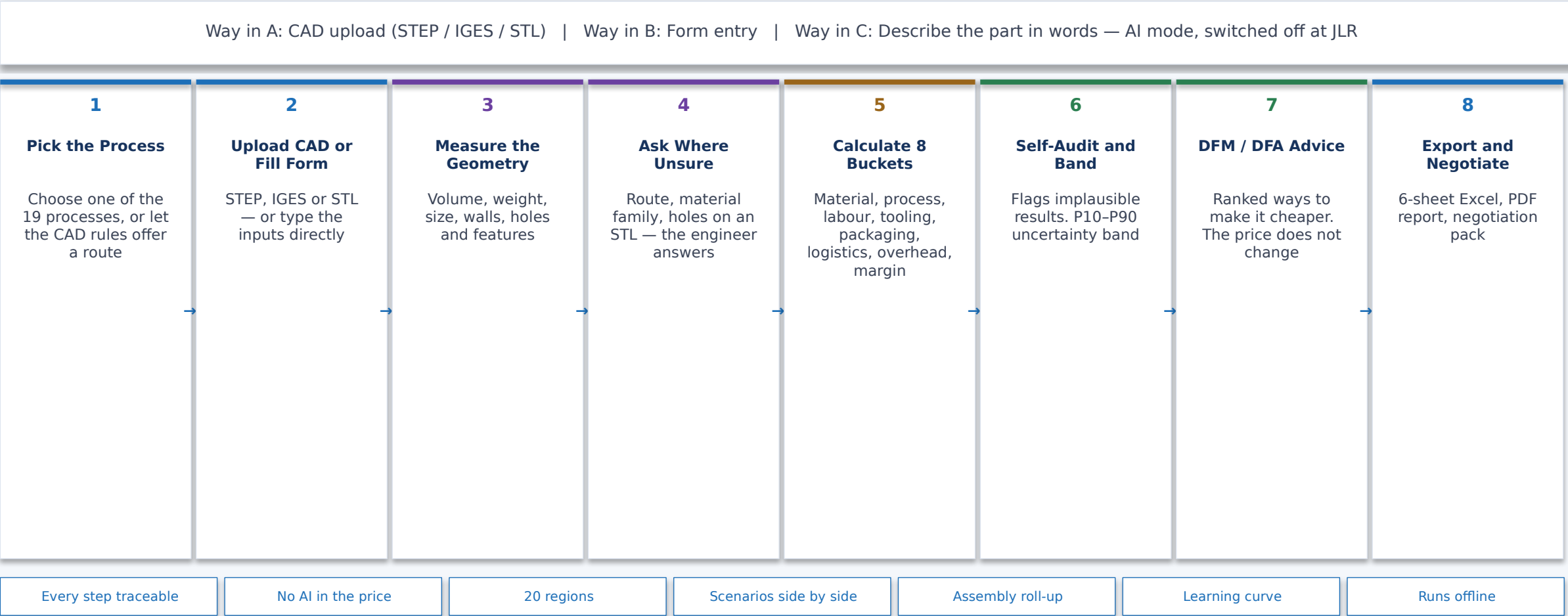
## The golden rule

No AI in the price. At JLR the whole chain runs on measurement and rules. Even where AI is switched on, it only reads and classifies. Every number above is arithmetic, bounded by the measured geometry and traceable to the rate library.

WORKFLOW

# From Part to Cost — Eight Steps

Two ways in at JLR: CAD upload or the form. The same engine either way.





FULL COVERAGE

# 19 Manufacturing Processes — One Engine

Plus an assembly roll-up and an automotive software cost model. The same eight cost buckets for all.

|                                                                                     |                                                                              |                                                                                        |                                                                                |                                                                                 |                                                                                  |                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <div>01</div> <div>CNC Machining</div> <div>Milling, turning, drilling</div>        | <div>02</div> <div>Casting</div> <div>HPDC, gravity, investment, sand</div>  | <div>03</div> <div>Cast + Machine</div> <div>Casting plus finish machining</div>       | <div>04</div> <div>Forging</div> <div>Hot die forging</div>                    | <div>05</div> <div>Sheet Metal</div> <div>Pressed: progressive / transfer</div> | <div>06</div> <div>Sheet Metal Fab</div> <div>Laser · punch · brake · weld</div> | <div>07</div> <div>Gear Cutting</div> <div>Hob · shape · skive · grind</div> |
| <div>08</div> <div>Injection Moulding</div> <div>Thermoplastics, multi-cavity</div> | <div>09</div> <div>Blow Moulding</div> <div>Hollow parts: tanks, ducts</div> | <div>10</div> <div>Extrusion</div> <div>Profile, pipe, tube</div>                      | <div>11</div> <div>Thermoforming</div> <div>Vacuum, pressure, twin-sheet</div> | <div>12</div> <div>Rotational Moulding</div> <div>Large hollow shapes</div>     | <div>13</div> <div>Rubber Moulding</div> <div>Compression, transfer, LSR</div>   | <div>14</div> <div>Composites</div> <div>Hand lay-up, RTM, autoclave</div>   |
| <div>15</div> <div>Painting / Coating</div> <div>E-coat, powder, wet, anodise</div> | <div>16</div> <div>BIW Assembly</div> <div>Spot weld, MIG, hem, bond</div>   | <div>17</div> <div>PCB Fabrication</div> <div>Bought-in price, fabricator tables</div> | <div>18</div> <div>PCBA</div> <div>SMT, through-hole, reflow</div>             | <div>19</div> <div>Wiring Harness</div> <div>Cut, strip, crimp, test</div>      | <div>20</div> <div>Assembly Roll-Up</div> <div>Multi-part BOM, one total</div>   | <div>21</div> <div>Automotive Software</div> <div>49-module cost model</div> |

ENGINEERING DEPTH

# What Sits Behind Every Number

Twelve layers, all of them arithmetic you can follow. Nothing is estimated by feel.

|                                                                                                                                        |                                                                                                                              |                                                                                                                                         |                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <p><b>Materials</b></p> <p>328 materials in the rate library dated 16 June 2026: metals, plastics, rubber, composites, laminates.</p>  | <p><b>Machines &amp; Labour</b></p> <p>178 machines and 42 labour grades, each rate editable per project and per region.</p> | <p><b>Process Routing</b></p> <p>The order of operations, and for each one: which machine, how long, who runs it.</p>                   | <p><b>Tooling &amp; NRE</b></p> <p>Dies, moulds and fixtures spread over annual volume × programme life.</p>              |
| <p><b>Yield &amp; Scrap</b></p> <p>What actually comes out good first time, driven by the features that make a part hard to build.</p> | <p><b>Regional Rates</b></p> <p>20 regions, each with its own labour, energy and machine rates.</p>                          | <p><b>Benchmarks &amp; Sensitivity</b></p> <p>Checked against typical cost ranges, and ranked by which inputs move the number most.</p> | <p><b>Learning Curve</b></p> <p>What the part costs as volume builds — the figure you need for a long-term agreement.</p> |
| <p><b>Quote Comparison</b></p> <p>Put the supplier's quote next to ours, bucket by bucket, in their currency or yours.</p>             | <p><b>Uncertainty Band</b></p> <p>A P10–P90 range on every result, from a Monte Carlo run over every cost driver.</p>        | <p><b>Gear Cutting</b></p> <p>Hobbing, shaping, skiving and grinding times worked out from the gear itself.</p>                         | <p><b>Landed Cost</b></p> <p>Duty, carbon levy and shipping terms — the cost at our door, not at the supplier's gate.</p> |

## ADVANCED CAPABILITIES

# Beyond a Cost Number

Five tools that turn the estimate into something you can negotiate with.

## Learning Curve

Cost falls as volume builds. Set the curve, read the cost at any year.

- Talk about year-three cost, not first-batch cost
- Built for long-term agreements and make-vs-buy

## Assembly Roll-Up

Cost a whole module, not just one part.

- Mix any of the 19 processes in one build
- Every component keeps its own cost model
- Body + paint + harness + PCBA = one total

## Quote Comparison

Log the supplier's quote. Any currency, converted for you.

- Compared with the should-cost, bucket by bucket
- The gap stated plainly: "Quote is X% above should-cost"

## What Moves the Number

Each driver flexed  $\pm 10\%$  (by default), ranked by impact.

- Shows the few inputs worth checking before you negotiate

## Scenarios Side by Side

Baseline, target and stretch on one screen.

- See exactly where the money moved
- Saved to your account to come back to

ADVICE LAYER

# DFM / DFA — How to Make the Part Cheaper

Runs after the cost is settled. It reads the result and advises — it never changes the price.

| Manufacturability Score (DFM)                                                                                                                                                                                                                                                | Assembly Score (DFA) | Actions                                                                                                                                                                                                                                                                                                      |                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1 - 10                                                                                                                                                                                                                                                                       | 1 - 10               | Ranked                                                                                                                                                                                                                                                                                                       | Score drops with each issue found · each action carries its own saving estimate |
| <div><div>What It Finds — Making the Part</div><div>Too much material going to scrap.<br/>Machines running well below what they should.<br/>Tooling, overhead or supplier margin out of line.<br/>Each issue graded critical, major or minor.</div></div>                    |                      | <div><div>What It Finds — Building It</div><div>Assembly done by hand that could be automated.<br/>Too many setups — each one adds cost and variation.<br/>Too many different fasteners, so tools keep changing.</div></div>                                                                                 |                                                                                 |
| <div><div>What to Do About It</div><div>A ranked list of actions, each with its own saving estimate:<br/><br/>Automate the manual operations<br/>Move to a near-net shape<br/>Raise machine uptime<br/>Plus quick wins: more volume, open-book overhead, re-bid.</div></div> |                      | <div><div>Checks on the 3D Model</div><div>A second layer reads the CAD model itself: casting, forging, moulding, machining, sheet metal.<br/><br/>It points at the faces that cause the problem.<br/>Every rule names its source.<br/>A finding is priced only when rates and volume are known.</div></div> |                                                                                 |

GLOBAL COVERAGE

# 20 Manufacturing Regions — Each in Its Own Currency

Pick a region and every rate resets together — labour, machines, energy, material and currency.

12 of the 20 regions — skilled labour, 2026 Q2

| Region   | Currency | Skilled labour (£/hr equiv.) | Where                  |
|----------|----------|------------------------------|------------------------|
| UK       | GBP      | £26.00                       | Base region            |
| Germany  | EUR      | £40.50                       | Western Europe         |
| Poland   | PLN      | £12.00                       | Central Europe         |
| Romania  | RON      | £7.50                        | Eastern Europe         |
| China    | CNY      | £7.90                        | East Asia              |
| India    | INR      | £5.10                        | South Asia             |
| Mexico   | MXN      | £7.50                        | Near-shore for the USA |
| USA      | USD      | £34.00                       | North America          |
| Thailand | THB      | £5.80                        | South-east Asia        |
| Vietnam  | VND      | £3.80                        | South-east Asia        |
| Brazil   | BRL      | £8.50                        | South America          |
| S. Korea | KRW      | £22.00                       | East Asia              |

Currency Follows the Region

Pick a region and the display currency switches to its own. Choose a currency by hand and it stays put.

Every Rate Resets

Eight labour rates, electricity, gas, machine rates and material prices — all set for the region.

Editable Rate Library

Version 2.1.0, dated 16 June 2026. Override any rate. JLR's own rate card converts in with a macro-free workbook.

All 20 Regions at Once

One table shows the same part costed in every region — a make-vs-buy view built on rates.

WHAT RUNS TODAY · WHAT WAITS

# Runs on Rules Today — AI Mode Waits for Approval

Left: what runs at JLR now, with AI switched off. Right: the optional AI features — built, but turned off.

## Runs Today — No AI

*Rules and arithmetic. Works offline on the laptop.*

### CAD to Cost

Measures the part, and asks you where the geometry cannot decide.

### Self-audit

Checks every result: wrong machine size, wrong volume for tooling, odd cycle times.

### Learning from actuals

Log a real quote or PO price. After 3 for a commodity, the band is corrected.

### Supplier price monitor

Compares logged prices with the should-cost and flags where a supplier sits above it.

## Optional AI Mode — Off at JLR

*Needs AI approval. Turned on by a setting, with no rebuild.*

### Describe a part

Type a description and the AI suggests the inputs. The engine still does the sum.

### PCB photo to BOM

Reads a board photo into a bill of materials, then costs it with the rules.

### AI assistant

Answers costing questions in plain English.

### AI agent

Works through a costing task step by step, when asked.

The common thread: AI never sets a price, on or off. Every number stays arithmetic you can check.

CHECKS AND PROOF

# The Tool Checks Itself — and Says What Is Proven

It checks every result before you see it, shows how unsure it is, and learns from real prices.

**It checks its own work**

A self-audit runs on every result. It flags a wrong machine size, tooling spread over the wrong volume, a material-only costing, implausible cycle times, and more.

**It shows how unsure it is**

Every result carries a P10–P90 band from a Monte Carlo run over every cost driver, including flat material prices.

**It learns from real prices**

Log a real quote or PO price with "Log Actual £". After 3 actuals for a commodity, the band is corrected by real data.

**What is proven — and what is not**

Status as of September 2026.

| Check               | Evidence                    | What it shows                                | Status     |
|---------------------|-----------------------------|----------------------------------------------|------------|
| Hand calculation    | Reference bracket, £24.34   | Engine arithmetic matches to <0.01%          | Proven     |
| Real-parts baseline | 6 production parts pinned   | A change that moves one fails the build      | In place   |
| Automated tests     | 2,438 tests + browser tests | Every commodity, exports, STL, accessibility | Every run  |
| JLR-paid prices     | None compared yet           | Measured as actuals are logged               | Not proven |

**Built to be checked, not just believed**

- 2,438 automated tests — if the logic breaks, the build fails
- Browser tests cost every commodity and check accessibility on every run
- The CAD engine is tested inside the image we ship

REAL PARTS · REGRESSION BASELINE

# Six Real Production Parts — Pinned in the Build

Measured from real CAD and costed by the product's own chain, at UK rates. Move one, and the build fails.

| Part                 | What it is             | Route (engineer's answer) | Should-cost | Measured                            | Why this route                                            |
|----------------------|------------------------|---------------------------|-------------|-------------------------------------|-----------------------------------------------------------|
| Steering knuckle     | Suspension upright, RH | Forging, steel            | £ 32.16     | 9.0 mm bulk wall · 6% fill          | Safety-critical, so forged; bore and faces machined after |
| Casting bracket      | Mounting bracket       | Cast + machine, steel     | £ 37.61     | 10.2 mm wall · 16% fill · 10 holes  | Named as a casting; holes and threads to finish           |
| PRCR002 housing      | Cast housing, 277 mm   | Cast + machine, aluminium | £ 52.67     | 15.1 mm bulk wall · 9% fill         | Assumed not pressure-tight — if it is, the cost rises     |
| Part1                | Machined part, 274 mm  | Machining, aluminium      | £ 38.93     | 10.4 mm bulk wall · 23% fill        | Machined from billet at this volume                       |
| Seat locking bracket | Pressed bracket        | Sheet metal, steel        | £ 4.96      | 1.55 mm wall · 15 bends · 4% fill   | The one genuine pressing                                  |
| Spur gear m3 z38     | 38 teeth, ISO class 8  | Gear cutting, steel       | £ 25.50     | Module 3, teeth counted off the tip | Routed by measuring the gear                              |

UK region, GBP | Answers are stated engineering judgements, so they can be argued with | This pins what the tool says — none of these has been compared with a price JLR paid



REPORTING & DEPLOYMENT

# Getting the Number Out — Safely

Reports people can act on, and a tool that is safe on a locked-down laptop.

### Excel — Six Sheets

The whole build-up, not just the total.

Summary · material · operations ·  
machine rates · labour rates ·  
and a traceability sheet.

### PDF Report and Negotiation Pack

The versions you send out.

Both carry the on-screen total to the penny.  
Files are named <type>-<part>-<date>.

### Windows Laptop Package

Copy a folder, double-click to start.

No installer, no admin rights, no internet.  
Listens on the laptop only, unless shared.

### Part Photos

Attach a photo to any calculation.

It appears on the cover of the PDF report,  
so everyone knows which part it is.

### Sign-In and Privacy

Sign-in on everything that holds data.

You see only your own saved scenarios.  
A password reset or "sign out everywhere"  
ends every old session.

### Help and Accessibility

Nobody needs training to start.

Getting started, a glossary and  
troubleshooting — all in the app.  
Every page passes WCAG 2.1 AA (automated).

VALUE — HONESTLY STATED

# What We Can Claim Today

Only what the repository can show. Accuracy against real prices is measured as they are logged.

|                                                                                                                                   |                                                                                                                                    |                                                                                                                                       |                                                                                                                  |                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <div>&lt;0.01%</div> <div>Engine vs hand calc, reference part</div>                                                               | <div>6</div> <div>Real parts pinned in the build</div>                                                                             | <div>2,438</div> <div>Automated tests</div>                                                                                           | <div>19</div> <div>Manufacturing processes</div>                                                                 | <div>0</div> <div>Estimates compared with a JLR-paid price, so far</div> |
| <div>Speed</div> <div>The engineer answers a few questions instead of building a spreadsheet. We will time it in the pilot.</div> | <div>Accuracy</div> <div>Arithmetic proven against a hand calc. Accuracy against real prices is measured as they are logged.</div> | <div>Negotiation</div> <div>A floor price you can defend, bucket by bucket. Supplier margin becomes visible instead of assumed.</div> | <div>Early Warning</div> <div>A cost from CAD at concept stage, while a change is still cheap.</div>             |                                                                          |
| <div>Where to Build It</div> <div>Compare 20 regions in one table. Make-vs-buy answered with real rates.</div>                    | <div>One Method</div> <div>The same model everywhere, with every assumption on record.</div>                                       | <div>The Right Volume</div> <div>See what the part costs as volume grows, and when a new process starts to pay.</div>                 | <div>Safe to Run</div> <div>A Windows folder on a locked-down laptop. No internet, no admin rights, no AI.</div> |                                                                          |

## VISION &amp; NEXT STEPS

# What Comes Next

CostVision runs today. Here is what could come next — and a small first step.

## Runs Today

- 19 manufacturing processes, plus assembly and software
- CAD to Cost on rules — asks where the geometry can't decide
- Rate library dated 16 June 2026, editable
- Self-audit and a P10-P90 band on every result
- Learns from logged quotes and PO prices
- 20 regions, each in its own currency
- Supplier quote comparison and negotiation pack
- Excel and PDF that match the screen to the penny
- Windows laptop package — no internet, no admin

## Next

Connect to Teamcenter and SAP  
Cost-target view for programme teams  
Compare several suppliers side by side

## Then

Calibrate from live ERP actuals  
Draft the RFQ from the should-cost  
Portfolio view for leadership

## Later

Cost that updates as the design changes  
Redesign suggestions from AI — only if AI is approved  
Supplier risk scoring

## Pilot — the Ask

- Two or three commodities, one programme team
- Four to six weeks, on real parts with real quotes and PO prices
- Log every actual, so accuracy is measured, not claimed
- Avinash Bhosale — Cost Engineering & Digital Innovation